

SRIJAN DAWADI

Software Engineer • Backend & Systems Focus

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PROFESSIONAL SUMMARY

Licensed Computer Engineer (NEC) and Software Engineer with production experience in backend architecture, modular REST API design, relational databases, and AI semantic search integration. Co-engineered and deployed a live marketplace platform (PerfectLink) in a 2-person engineering team, architecting core Django REST Framework backends, PostgreSQL schemas, and a vector similarity search engine (all-MiniLM-L6-v2 + pgvector). Engineered and delivered a desktop computer vision management system to a paying client in production.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, HTML, CSS

Backend & APIs: Django, Django REST Framework (DRF), REST APIs, JWT Authentication, Django ORM, Serializers, ViewSets, Modular Architecture, API Design

Databases & Search: PostgreSQL, pgvector, Vector Embeddings, Semantic Search, Database Migrations

Frontend & UI: Next.js, React.js, Tailwind CSS

AI & Computer Vision: all-MiniLM-L6-v2, Embeddings, InsightFace, ONNX Runtime, Passive Liveness Detection, Scikit-learn, TensorFlow, Autoencoders, NLP

Tools & Workflow: Git, GitHub, OpenStreetMap API Integration

PROFESSIONAL EXPERIENCE

Software Engineer — PerfectLink (*perfectlink.com.np*)

Aug 2025 – Present

- Co-engineered and deployed a live service and product marketplace platform for Nepal as part of a 2-person IT team.
- Architected an AI semantic search engine using all-MiniLM-L6-v2 embeddings and pgvector on PostgreSQL, matching intent queries like "fix my pipe" to relevant "plumber" listings.
- Built decoupled Django REST Framework backend modules for accounts, listings, and payment workflows using JWT authentication, ViewSets, serializers, pagination, and search filtering.
- Developed frontend user interfaces and seller dashboards in Next.js, TypeScript, and Tailwind CSS while integrating OpenStreetMap location features.
- Refactored backend architecture to maintain clean modular code separation and executed production PostgreSQL database migrations.

Freelance Software Developer — Smart Gym & Cafe Management System

Apr 2026 – Jul 2026

- Designed, built, and delivered an end-to-end desktop management system to a paying gym client, replacing manual paper records.
- Integrated InsightFace and ONNX Runtime to build a facial-recognition attendance module with passive liveness detection to prevent photo spoofing.
- Developed Django and PostgreSQL modules for role-based access control, payment tracking, membership expiry alerts, and cafe inventory.
- Deployed and configured the production system on client hardware where it currently processes 100+ daily member check-ins.

KEY ENGINEERING PROJECTS

Network Anomaly Detection System | *Python, TensorFlow, Autoencoder*

Personal Project

- Engineered an unsupervised anomaly detection model using TensorFlow Autoencoders trained exclusively on benign network traffic from the NSL-KDD dataset.
- Implemented reconstruction error thresholding to flag zero-day intrusion patterns without labeled attack data, achieving 97%–99% detection accuracy on test traffic.

E-Commerce Recommendation System | *Python, Scikit-learn, NLP*

Personal Project

- Built a content-based recommendation system evaluating 10,000 Kaggle product descriptions using text preprocessing, Bag-of-Words vectorization, and cosine similarity ranking.

EDUCATION & PROFESSIONAL CREDENTIALS

Licensed Computer Engineer — Nepal Engineering Council (NEC)

Passed Exam: July 2026

Bachelor of Computer Engineering — United Technical College

2022 – 2026

CGPA: 3.76 / 4.00 • Standing: Dean's List • Completed Coursework: March 2026